

Painting Portals: connecting homes through live paintings

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Figure 1: Two Painting Portals incorporated into the decor of a home as framed abstract paintings that slowly change over time using the camera stream from a remote location.

ABSTRACT

Feeling co-present and connected over distance is a challenge. To create a sense of sharing a space over distance and being together in the periphery of daily life, we designed and implemented a pair of connected picture frames that display paintings generated from live camera streams. These paintings are abstracted based on the distance of the user from the device, and are sent between two portals that can be placed anywhere in the world. We deployed the Painting Portals during a week-long field study investigating how a distance-separated mother and daughter experienced the use of the Painting Portals. We generated recommendations for future design and research, including: (1) stylizing the paintings in a way that better captures emotion/sentiment, (2) placing multiple Painting Portals in different common areas throughout a home, (3) incorporating a video call feature into the Painting Portals, and (4) adding the option to turn off one's camera for more privacy.

CCS CONCEPTS

• **Human-centered computing** → **Ubiquitous and mobile devices**.

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1 INTRODUCTION

When families are separated by distance, it can be challenging to establish and maintain the same level of connection as in a co-located household. Many distance-separated families and couples in long-distance relationships are faced with this challenge, especially now during the COVID-19 pandemic. Existing communication technologies - such as video calls [8, 10–12] - revolve around high-engagement interaction sessions and shared activities. However, a major part of the interactions people share in a co-located household are implicit, passive actions that span long periods of time rather than session-based interactions. Examples of this are hearing someone walking around in another room in the house, or seeing them in your peripheral vision. Previous research has suggested that passive interaction can create a heightened sense of closeness [4].

As a solution for supporting passive awareness of a distance-separated household, we propose *Painting Portals* (Figure 1). Inspired by the moving paintings of the Harry Potter universe, Painting Portals are a series of wall-hanging digital paintings that provide live camera streams between two households. To mitigate possible privacy concerns and to add to the decor of the home, these camera streams are stylized as paintings that are more detailed when the remote user approaches the frame (indicating the intention to interact) and are more abstract when they are far away.

Our contributions include the design and implementation of a novel technology that can be incorporated into distance-separated households as an abstract painting, and a week-long field study confirming the value of the Painting Portals in a real-world context at an early design stage in order to assess how the design needs to be adapted in the future. Our recommendations include (1) stylizing the paintings in a way that better captures emotion/sentiment (2) placing multiple Painting Portals in different common areas throughout a home (3) incorporating a video call feature into the Painting Portals (4) adding the option to turn off one's camera for more privacy.

2 RELATED WORK

2.1 Video streams

Research has shown that video conferencing in the home can be a valuable method of facilitating connections [5, 9, 10] between family members. Another study showed that video calls created a sense of "closeness" [12]. However, these studies also revealed that video conferencing poses challenges with positioning the camera to provide good views of activities, especially when this requires movement between rooms, or when young children are involved [5, 10, 12]. Other modes of connecting over distance - such as touch [21, 23, 24] - are important as well, but we hoped to address these pitfalls of video systems, especially given the popularity of video as a mode of connection.

Always-on video is an uninterrupted video stream that is not turned off after a conversation, but is always left on, creating a sense of shared life and connection [15]. Hutchinson et al created videoProbe, a system that has an always-on camera, but does not stream video [9]. Instead, the videoProbe system snaps images when they are held steady for three seconds, sending them to connected devices. Judge et al investigated a design that combined the use of always-on video with a messaging system to create portals between three homes [11]. In order to preserve privacy, this system did not have an audio connection. The authors found that for some users, this system led them to behave as though they lived in the same house, while other users found the constant sharing and communication overwhelming.

While these prior works lay the fundamental groundwork for domestic video communication research, they also present an opportunity for further exploration. None of the prior systems focus on peripheral, passive, unconscious social awareness and togetherness. Previous research primarily explores video systems that require conscious, active engagement. Judge et al propose an always-on connection that could be partially ignored, though in practice it did not effectively blend into the background and created some pressure to interact. In this project, designed a system that blends into the home environment in a way that does not require the user to move the camera or interact in any way.

2.2 Integrated home displays

When we are co-located with members of our household, much of our social awareness is in the periphery, and is always present. For example, a study by de Ruyter and Huijnen found that the peripheral awareness of a remote group of friends' video feeds resulted in feelings of social presence and belonging while watching

a television program together [4]. This phenomenon was confirmed by a secondary study by Markopoulos et al [13], and was leveraged in several social presence systems. For example, the Social Hue is a system that uses lighting to create social presence between the elderly and their caregivers by adjusting the lighting according to users' physiological information [3]. Another example is Damage, a wearable prototype display that shows ambient group messages [25].

For a device in the home to effectively blend into the periphery, it must be designed as an object that fits naturally in the space. With prior research showing that a major difficulty with telepresence technology is attending to devices and positioning them around the home [10–12], finding ways of integrating technology in the home unobtrusively is important. Examples of prior displays that have been integrated into the home as common objects include a couch that uses lights and haptics to display emotional states [14], a set of drink coasters that record and replay a family's past moments [22], an interactive window display that can display animations and messages [1], an ambient frame display for elder care [2], and HomeNote, a wall-mounted picture-frame-like display to support home-to-home messaging [19]. Another past system that takes the approach of in-home decor is a set of flowers that recognize touch input and light up with LEDs [24]. These past designs show the importance of using familiar form factors for in-home technology in order to smoothly integrate the systems into existing spaces. We draw from this to inform the design of Painting Portals as picture frames that can be placed throughout the home as decor. With the exception of the Social Hue [3] and HomeNote [19], these past technologies do not support interactions between multiple homes. The Social Hue uses lighting to create social awareness, and HomeNote uses text messaging. We branch out by using image data to capture the remote space and household members in more depth in order to create a sense of co-presence.

2.3 Slow technology

Slow technology is a design framework shifts away from the model of quick fleeting interactions, allowing for interactions to be experienced over months or years [16]. Hallnas et al. describe the potential of slow technology to support reflection, rest, and expression, and they emphasize the importance of aesthetics in slow technology [6]. An example of this is PhotoBox, a photo printer that is designed to fit within the aesthetic of the home that automatically prints photos at random times over several months [16]. Another example is LunchTime, an interactive multiplayer game about making healthy food choices, where each game is played across 10 days, with each move happening once every 12 hours [17]. Hallnas and Redstrom explain that slow art can be a method of amplification in the way that it brings emphasis to things that may otherwise have quickly passed by. Meanwhile, it also creates abstraction [7]. These are two qualities of slow technology that we are inspired by, and that we apply to the space of slow home-to-home interactions that are richly social while remaining private and abstract.

3 DESIGN

Painting Portals are pairs of framed digital paintings that provide live camera streams between two households. These paintings are

automatically generated from live captured images and sent between the devices. Each frame displays the painting generated from the image captured by the opposite frame. Following the principles of slow technology [6], these paintings, while “live”, are meant to update slowly, so that they give the impression of being static paintings in the home. To achieve this impression, the displayed image is updated every 10-30 seconds in-between the painting changes. The goal of Painting Portals is to create feelings of sharing a space over distance and being together in the periphery of daily life. Consider the following scenario:

Aragorn and Arwen are in a long-distance relationship, as their jobs are on opposite sides of the continent. They started using Painting Portals to bring back the feeling of togetherness that they experienced when living in the same house. When Aragorn passes through his living room, he sees an abstracted painting of Arwen’s living room, and feels like their living spaces are connected. He goes to the painting in the kitchen, and notices that Arwen is in her kitchen as well.

To mitigate possible privacy concerns and to add to the decor of the home, these paintings are abstracted as geometric shapes. They are more detailed when the remote user approaches the frame (indicating the intention to interact) and are more abstract when they are far away.

We created the hardware of the Painting Portals (Figure 1) by using two computer monitors (11x13in and 17x11in) and removing their stands so that they would appear more like picture frames. We then built wooden frames for the monitors to be placed in, so that they could be optionally hung up on walls and so that they look less like monitors and more like picture frames that one might find in a home.

4 IMPLEMENTATION

A 2MP camera integrated into each frame captures live image data from the frame. Next, distance data is recorded and the image is accordingly stylized as a painting. To update the level of painting abstraction based on the user’s distance from the Painting Portal, we used an Arduino micro-controller with an ultrasonic transducer. This distance information then determines the level of detail in the painting, with more detail showing as the user gets closer. Our goal was to ensure that this applies only for intentional interactions with the device, leaving the painting highly abstract as the inhabitants of the house go about their lives. Thus, we chose to increase detail starting at a distance of 40 cm, which is likely to take place during an intentional interaction. The distance readings from the Arduino are written to a serial port as bytes. These values are then retrieved by Processing and used for abstraction.

We stylized the paintings using code adapted from [18]. This process involves random forest regression, an artificial intelligence method that involves a “forest” of trees, which each branch to a specified depth. Given a dataset, each tree splits the dataset into related clusters recursively until the specified depth is achieved. This process is non-deterministic, but as the number of trees increases the model begins to converge towards a common result. We chose to use one tree to create a visually interesting and more variable result, as we predicted that the captured images may stay the same throughout large portions of the day. The depth of the tree indicates the level of detail, with more depth resulting in higher

detail. At a range of closer than 40 cm, the depth of the regressor is correspondingly mapped between 8 and 20. In Figure 2 we see this progression.

This painting is then uploaded to an online repository, and the corresponding painting from the other device is downloaded from the same repository, resized, and displayed. For the two Painting Portals to display paintings generated by one another, we implemented a networking component. We used the Google Drive Java API, assigning a Google Drive folder as a repository that both devices could upload to and download images from, regardless of where in the world they are located. This file transfer takes approximately 10-15 seconds for upload and download altogether. A benefit of this approach is that the system can be very easily expanded to work with multiple devices, as long as those devices have the API credentials as well. A reason for including multiple Painting Portals may be that a household may wish to have several rooms that are connected to their corresponding spaces in the distant household. For example, they may have portals between their kitchens, hallways, and living rooms.

5 PRELIMINARY STUDY

We conducted a week-long preliminary field deployment of the Painting Portals in order to investigate their use in two distance-separated homes. With the Painting Portals being in an early design stage, we hoped to gain insight into whether the design is worth pursuing in more depth in future research. At this stage in the design process, we decided that a small formative two person study would be the most appropriate. It provided us with the opportunity to see how the Painting Portals are used in a real setting so that we can improve them before committing to a larger summative evaluation. The study was approved by our research ethics board. Here, we present our methods and results.

5.1 Method

Our participants included a mother (age 51) and adult daughter (age 20) who live in separate homes approximately 40 kilometers apart. Both participants had a Painting Portal in their homes throughout the week, placed among their decor. The mother lives in a large two-bedroom apartment, and chose to place the frame above the fireplace in the living room. The daughter on the other hand, lives alone in a small studio apartment in a university student residence, and placed the frame on her desk. Both homes had high-speed internet connections. Aside from the researcher, there were no other members in either household during the study.

As both participants were close family members of one of the researchers, the researcher was able to spend the first half of the week in the daughter’s home and the second half in the mother’s home in order to observe their interactions and reactions first hand and record notes. At the end of the week, we conducted an interview with each participant in order to learn more about their experiences. Our research questions were the following:

- *RQ1:* How are the participants’ senses of togetherness and co-presence impacted by the use of the Painting Portals?
- *RQ2:* How do the Painting Portals play a role in participants’ daily routines? How do they interact passively and actively?



Figure 2: The progression of distance-based abstraction, from farthest (most abstract) to closest (most detailed) distances from the sensor.

- *RQ3*: How are the participants' feelings of privacy impacted by the use of the Painting Portals?

5.2 Results

5.2.1 Connection and Presence. Both participants reported increased feelings of connection with one another, as well as a sense of presence. In the interview, the mother explained that the sense of togetherness was not impacted by the abstraction: "I could feel that she was there, even though I couldn't see her face completely".

The daughter described in the interview, "It just felt like we were living in the same house [...] I think it sort of expanded both houses to be one". Both participants explained in the interview that they felt that the other participant was *more* present in their own home than they were in the other home. This may have been a result of the limited view into the other home. The mother described, "I felt like the parts I could see, I could feel my presence, but I couldn't see the whole thing".

The mother elaborated that she particularly appreciated the fact that she could "check-in" on her daughter just as she could if they were living in the same home. We observed that on a stormy night during the study when the daughter was out late, the mother noted that she felt relieved that she could just check to see if the daughter had made it home safely without sending a text message and waiting for a response.

5.2.2 Passive and Active Interactions. Participants said that they both found the Painting Portals to fall into the background of their attention throughout the week. We observed that each day, they checked in on the painting for updates around 3-6 times. The mother explained in the interview that she checked in on the system "when I was thinking about her". It is important to note that the number of check-ins may have been influenced also by the fact that the mother worked outside the home during the day, leaving the home empty and resulting in a somewhat static image.

The daughter explained that she enjoyed the fact that the Painting Portal was in the background of her attention, as she found that it made her feel more comfortable. She contrasted this with the level of self-consciousness caused by seeing one's own image in a video call. She explained that the conscious awareness that she experiences in traditional video calls causes her to pay more attention to being on camera, which can feel uncomfortable at times. With the Painting Portal taking up less of her conscious attention, she could more easily forget about it and feel more at ease.

Ordinarily, the two participants would share a daily phone call (around 5 minutes long). During this study, we observed that these phone calls grew longer and more engaging than normal (15-20 minutes). While speaking to each other, both sat in front of their Painting Portal, which adjusted to show higher-detail images as they spoke, almost like a video call. Unlike a video call, we noticed that the interactions were somewhat more playful. For example, they made funny poses and expressions and waited for it to update on the other side to see if the other participant could guess what they were doing.

5.2.3 Selective Detail. The adjustable level of abstraction (based on distance) was a feature that we observed the participants using during the week, and that they mentioned in the interview that they appreciated. The mother explained that the high level of detail created more of a connection, saying, "when she was close to the screen, I could see her more so I could feel her more".

In the interview, both the mother and daughter suggested alternative non-distance based methods of detail control. The daughter, who was living in a small space, had a harder time managing her distance from the screen, and felt that she should avoid her desk unless she wants to interact with the Painting Portal. The mother on the other hand, wanted to be able to control the level of detail when she was farther away, describing a situation where she could be in the kitchen (which has a view of the living room) but still interacting with the Painting Portal a few meters away.

An interesting interaction that took place over this week was between the daughter and the pet fish she had left behind in her mother's home when she moved away. During the day when the mother was not home, we noticed that the daughter would still check in on the Painting Portal to see if she could spot the fish through the abstraction, and expressed that she wished that the fish specifically could be less abstracted. On one of the days, the mother had slightly shifted the angle of the Painting Portal before she left the home, which left the fish tank out of view. This resulted in the daughter asking the mother to shift it back the next day so that she could watch the fish tank. In the interview, the daughter expanded on this, saying, "it should have been closer to his tank, because I know he's smart enough that he would recognize me too if he could see me up close and he would be less bored and lonely".

5.2.4 Privacy. Both participants expressed in the interview that the abstraction made them feel more comfortable in terms of privacy, but the mother and daughter had very different experiences with privacy in general when using the Painting Portals. The mother, who said she wanted to be connected with her daughter, reported feeling "completely comfortable" with having abstracted images sent. The daughter on the other hand, expressed that she is generally uncomfortable with the idea of being in front of a camera, saying, "I hate being on camera, I hate having someone seeing me from unflattering angles all the time". She explained that while she was aware that her image was being abstracted, she could not see what her own image looked like and therefore did not fully trust what was being sent. She also expressed that having recently moved out of her parents' home for the first time, she valued her independence and wanted to be less connected at times. "I felt a little too connected - the idea of someone constantly being there makes me feel overwhelmed with their presence".

This may have stemmed in part from the fact that she lives in a studio, where nearly her entire home was in view of the camera. She explained, "maybe if I lived in a multi-room house it would feel like we were living in the same house, but it felt like we were living in the same room, like she was in my bedroom as well."

6 DISCUSSION AND FUTURE WORK

The design and evaluation of the Painting Portals led us to new insights and implications for future research. We discuss our findings and recommendations here.

RQ1: How are the participants' senses of togetherness and co-presence impacted by the use of the Painting Portals?

The study revealed that using the Painting Portals made the participants feel more connected, mainly due to the way that it seemed to "expand" both of the spaces to feel like one larger shared space that they both lived in. They especially felt that the other participant was present in their home alongside them, which helped to combat feelings of isolation and loneliness.

The mother felt most connected when her daughter was close to the camera, as she could "feel" her daughter more when the painting had a higher level of detail. This suggests that while abstraction is a useful privacy measure, it may result in some loss of emotional connection. Future work should explore stylization methods that capture sentiment and expressions while still maintaining privacy.

Just as an artist may draw attention to salient areas in a painting through the use of colour or detail [20], the Painting Portals could also make use of artistic techniques to create more of an emotional connection.

A limitation of this study was that each home only had one Painting Portal placed in a single room, rather than multiple Painting Portals placed throughout the home. Future research could investigate how this plays a role in the feelings of sharing a space.

RQ2: How do the Painting Portals play a role in participants' daily routines? How do they interact passively and actively?

For the most part, the participants found that the Painting Portals fell into the background of the daily routines, as they mainly paid conscious attention to it when they were thinking of the other person.

The participants actively used the Painting Portals when they were sharing phone conversations, like a very slow, abstracted video call. While the slowness and abstraction supported a sense of playfulness, this may have been a result of the novelty of the technology during this week of use. It would be interesting for future research to investigate the affordances of this method of interaction, and potentially consider incorporating a video call feature directly into the Painting Portals.

RQ3: How are the participants' feelings of privacy impacted by the use of the Painting Portals?

While the abstraction of the paintings made the mother feel comfortable with having her images sent, it did not fully address the daughter's privacy concerns. Living in a small space, the daughter felt overly exposed and conscious of always being on camera. This suggests that Painting Portals could be best suited to being placed in the more "common" areas of the home - those that would typically be shared spaces - rather than in private spaces such as the bedroom.

Additionally, while the Painting Portals are meant to be always-on, it may be important to be able to turn it off when a user does not feel comfortable with being on camera. Future design should explore how the design of the frames could incorporate a physical cover for the camera in order to create a sense of trust.

7 CONCLUSION

Feeling co-present and connected over distance is a challenge. To create feelings of sharing a space over distance and being together in the periphery of daily life, we designed and implemented a pair of connected picture frames that display paintings generated from live camera streams. These paintings are abstracted based on the distance of the user from the device, and are sent between two portals that can be placed anywhere in the world. We deployed the Painting Portals during a week-long field study evaluating how a distance-separated mother and daughter experienced the use of the Painting Portals, and generated recommendations for future design and research.

Limitations of this study include the small sample size (two participants) and relatively short study duration (one week). While this was appropriate for an early design stage and gave us lots of initial insight into the viability of the Painting Portals design, the results

do not generalize to long-term interactions with these devices or, for example, different relationships between the members of the remote households. Additionally, the researcher conducting the study was a family member of both participants and was physically present during the study, which may have led to biased results. That said, our results point to both positives and negatives with the technology which illustrates our participants were indeed being critical of the design ideas. Future research should consider expanding the studied population and increasing the duration of the study to overcome these limitations.

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